

$$\sigma = E \cdot \epsilon$$

$\sigma \rightarrow$ Tensão (força aplicada)

$E \rightarrow$ módulo de elasticidade / módulo de Young $\rightarrow$ Cada material possui o seu.

$\epsilon \rightarrow$ Deformação linear.

$$\sigma = \frac{F}{A} \rightarrow \text{Força}$$

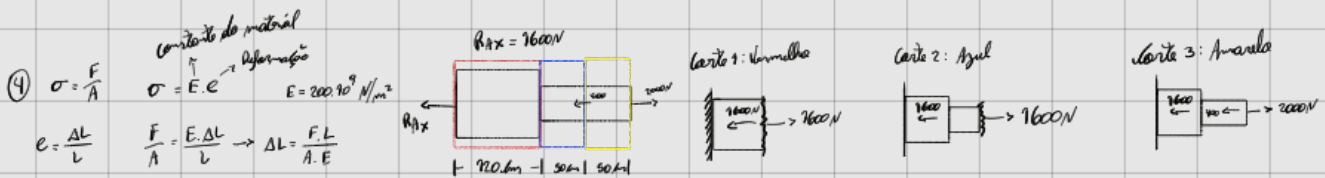
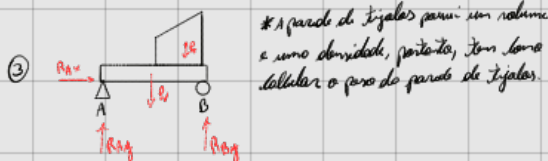
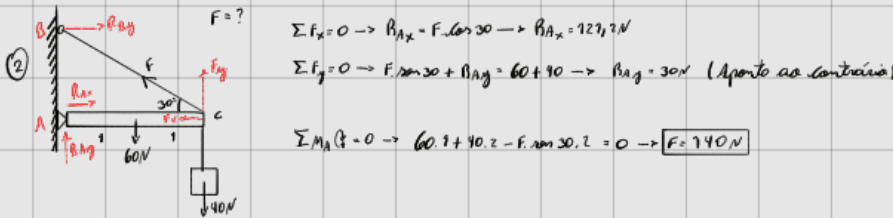
$\rightarrow$ Área do eixo transversal

$$\epsilon = \frac{\Delta L}{L} \rightarrow \text{Deslocamento}$$

$\rightarrow$ Comprimento

$$\Rightarrow \text{Sabendo que } \sigma = E \cdot \epsilon \therefore \Delta L = \frac{F \cdot L}{A \cdot E}$$

$\rightarrow$ Deslocamento $\rightarrow$ Comprimento



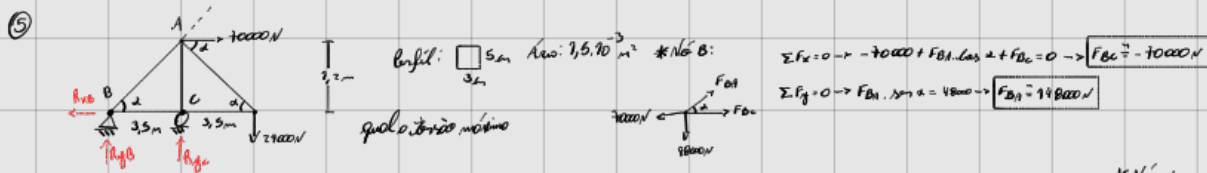
$$A_1 = 0,282 \text{ m}^2 \quad A_2 = 0,0406 \text{ m}^2$$

* Após fazer as cortes para análise de força precisa sempre de:

$$\Delta L = \frac{1}{200 \cdot 10^9} \left(\frac{1600 \cdot 3}{0,282} + \frac{1600 \cdot 0,5}{0,0406} + \frac{2000 \cdot 0,5}{0,0406} \right) = 1,67 \cdot 10^{-3} \text{ m}$$

* Corte: O que está passando aqui direito?

* Lei de Hook: Quanto mais alongo?



$$\alpha = \arctg \left(\frac{2,2}{3,5} \right) = 38,9^\circ$$

$$\Sigma F_x = 0 \rightarrow R_{Ax} = 10000 N$$

$$\Sigma F_y = 0 \rightarrow R_{Ay} + R_{Dy} - 20000 = 0 \rightarrow R_{Dy} = -48000 N$$

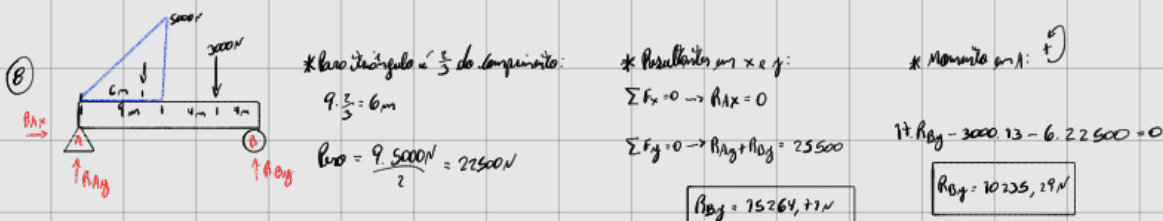
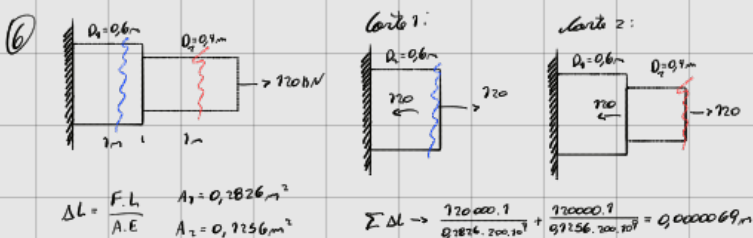
$$\Sigma M_D = 0 \rightarrow 20000 \cdot 7 + 10000 \cdot 12 - 3,5 \cdot R_{Ay} = 0 \rightarrow R_{Ay} = 70000 N$$

$$\sigma_{máx} = \frac{F_{máx}}{A_{máx}} \rightarrow \sigma_{máx} = \frac{148000}{1,5 \cdot 10^{-3}} = 98,6 \text{ MPa}$$

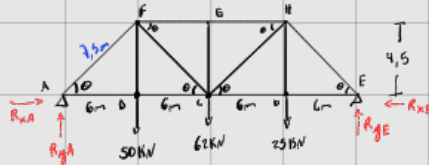
* Retorno: - Fazer o diagrama de força bin, calcular momentos e deslocar valores

- Com esses valores deslocados fazer o análise modal e deslocar forças

- Retornar o tensão máxima aplicada em uma das partes do material.



$$\sin \theta = \frac{4,5}{7,5} = \frac{3}{5} \quad \cos \theta = \frac{6}{7,5} = \frac{4}{5}$$



$$\sum F_x = 0 \rightarrow R_{xE} - R_{xA} = 0 \rightarrow R_{xA} = 0$$

$$\sum F_y = 0 \rightarrow 50 + 62 + 25 = R_{yA} + R_{yE} \rightarrow 137 - 62,25 = R_{yA} \rightarrow R_{yA} = 74,75 \text{ kN}$$

$$\sum M_A = 0 \rightarrow -50 \cdot 6 - 62 \cdot 12 - 25 \cdot 18 + R_{yE} \cdot 24 = 0$$

$$24 R_{yE} = 1998 \rightarrow R_{yE} = 83,25 \text{ kN}$$

$$\text{No A: } F_{AH} \cdot \sin \theta + R_{yA} = 0$$

$$F_{AH} \cdot \frac{3}{5} = 74,75 \rightarrow F_{AH} = 124,58 \text{ kN}$$

$$F_{AF} \cdot \cos \theta + F_{AB} = 0 \rightarrow F_{AB} = 99,264 \text{ kN}$$

$$\text{No B:}$$

$$\sum F_y = 0 \rightarrow F_{BH} - 50 = 0$$

$$F_{BH} = 50 \text{ kN}$$

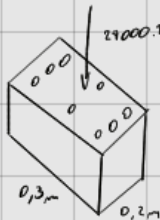
$$\sum F_x = 0 \rightarrow F_{BC} - F_{BH} = 0$$

$$F_{BC} = F_{AB} \rightarrow F_{BC} = 99,264 \text{ kN}$$

$$\text{No F: } \sum F_y = 0 \rightarrow F_{AF} \cdot \sin \theta - F_{FC} \cdot \sin \theta - 50 = 0$$

$$F_{FC} = 24,44 \cdot \frac{5}{3} = 40,73$$

$$\sum F_x = 0 \rightarrow -F_{FH} \cdot \cos \theta + F_{FC}$$



$$E_{Aço} = 200 \cdot 10^9 \text{ Pa}$$

$$E_{concreto} = 20 \cdot 10^9 \text{ Pa}$$

$$8 \text{ barras de Aço}$$

$$P_c = 2000 \cdot 10^3 \text{ N}$$

$$P_A = 400 \cdot 10^3 \text{ N}$$

$$\text{Deformação}$$

$$\sigma = E \cdot \epsilon \rightarrow \epsilon = \frac{\sigma}{E} = \frac{F}{A \cdot E}$$

$$\sigma = \frac{F}{A}$$

$$\epsilon = \frac{\Delta L}{L}$$

$$A_{Aço} = 8 \cdot \pi \cdot r^2$$

$$A_{con} = 0,06 - 8 \cdot \pi \cdot r^2$$

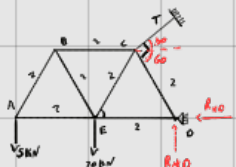
$$1,17 \cdot 10^{-3} = 8 \cdot \pi \cdot r^2$$

$$\text{Como a seção, as deformações precisam ser iguais:}$$

$$\epsilon_{Aço} = \epsilon_{concreto} \rightarrow \frac{400 \cdot 10^3}{A_{Aço} \cdot 200 \cdot 10^9} = \frac{2000 \cdot 10^3}{(A_c - 8 \cdot \pi \cdot r^2) \cdot 20 \cdot 10^9}$$

$$50 \cdot A_{Aço} = 0,06 - A_{Aço}$$

$$51 A_{Aço} = 0,06 \rightarrow A_{Aço} = 1,17 \cdot 10^{-3}$$



$$\sum M_D = 0$$

$$-5 \cdot 4 - 20 \cdot 2 + 2 \cdot T = 0$$

$$T = 30 \text{ kN}$$

$$T_H = T \cdot \cos 30 = 25,98 \text{ kN}$$

$$T_V = T \cdot \sin 30 = 15 \text{ kN}$$

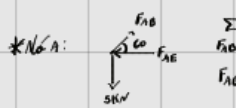
$$\sum F_x = 0$$

$$R_{xD} = 25,98 \text{ kN}$$

$$\sum F_y = 0$$

$$15 + R_{yD} - 5 - 20 = 0$$

$$R_{yD} = 10 \text{ kN}$$



$$\sum F_x = 0$$

$$F_{AB} \cdot \sin 60 = 5$$

$$F_{AB} = 5,773 \text{ kN}$$

$$F_{AE} = 5,773 \cdot \cos 60$$

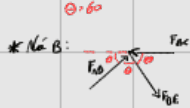
$$F_{AE} = 2,8865 \text{ kN}$$

$$\sum F_y = 0$$

$$-F_{BE} + F_{AB} \cdot \cos 60 = 0$$

$$F_{BE} = 5,773 \cdot \cos 60$$

$$F_{BE} = 2,8865 \text{ kN}$$



$$\sum F_x = 0$$

$$F_{AB} \cdot \cos 60 + F_{BC} - F_{BE} \cdot \cos 60 = 0$$

$$2 \cdot 5,773 \cdot \cos 60 + F_{BC} = 0$$

$$F_{BC} = -5,773 \text{ kN}$$

$$\sum F_y = 0$$

$$F_{AB} \cdot \sin 60 - F_{BE} \cdot \sin 60 = 0$$

$$F_{AB} = F_{BE} = 5,773 \text{ kN}$$



$$\sum F_x = 0$$

$$R_{yD} - F_{CD} \cdot \sin 60 = 0$$

$$F_{CD} = \frac{R_{yD}}{\sin 60}$$

$$F_{CD} = 11,547 \text{ kN}$$

$$\sum F_y = 0$$

$$F_{CD} \cdot \cos 60 + F_{DE} - R_{xD} = 0$$

$$11,547 \cdot \cos 60 + F_{DE} - 25,98 = 0$$

$$F_{DE} = 20 \text{ kN}$$

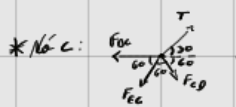


$$\sum F_x = 0$$

$$F_{EC} \cdot \cos 60 + F_{DE} - F_{BE} \cdot \cos 60 - F_{AE} = 0$$

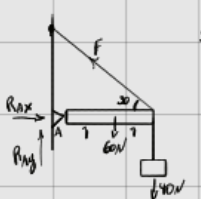
$$F_{EC} \cdot \cos 60 = 14,927$$

$$F_{EC} = 28,86 \text{ kN}$$



$$\text{Não preciso fazer mais, pois já determinei todas as forças}$$

$$\sigma_{max} = \frac{F_{max}}{A} \rightarrow \sigma = \frac{28,86 \cdot 10^3}{9,008} = 3607500 \text{ Pa}$$



$$\text{Somação de forças:}$$

$$\sum F_x = 0 \rightarrow R_{Ax} = F \cdot \cos 30$$

$$R_{Ax} = 127,2 \text{ N}$$

$$\sum F_y = 0 \rightarrow -100 + R_{Ay} + F \cdot \sin 30$$

$$R_{Ay} = 30$$

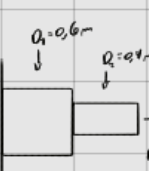
$$\text{Momento no ponto A:}$$

$$\sum M_A = 0$$

$$402 + 60 - F \cdot \sin 30 = 0$$

$$F = \frac{100}{\sin 30} = \frac{200}{1} = 200 \text{ N}$$

$$F = 740 \text{ N}$$



$$A_1 = \pi \cdot (0,3)^2 = 0,2827 \text{ m}^2$$

$$A_2 = \pi \cdot (0,2)^2 = 0,1257 \text{ m}^2$$

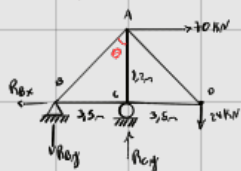
$$\text{Alongamento: } \Delta L = \frac{F \cdot L}{E \cdot A} \rightarrow \Delta L = \frac{1}{E} \cdot \left(\frac{F \cdot L}{A} \right)$$

$$\rightarrow \Delta L = \frac{1}{200 \cdot 10^9} \cdot \left(\frac{120 \cdot 10^3 \cdot 1}{0,2827} + \frac{120 \cdot 10^3 \cdot 1}{0,1257} \right)$$

$$\Delta L = 7,02 \cdot 10^{-6}$$

$$\text{Como tem duas regiões e não há forças no meio do larro, vamos fazer dois larros}$$

angulo $\frac{3,5}{1,2} \rightarrow \theta = 71^\circ$ $\alpha = 18,9^\circ$



$$\Sigma F_x = 0 \rightarrow R_{Bx} = 70 \text{ kN}$$

$\rightarrow$ Vale para ambas.

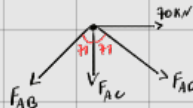
$$\Sigma F_y = 0 \rightarrow R_{By} + R_{Cy} = 24 \rightarrow R_{By} = -48 \text{ kN}$$

$$\Sigma M_B = 0 \rightarrow -R_{Cy} \cdot 3,5 + 24 \cdot 7 + 70 \cdot 1,2 = 0$$

$$R_{Cy} = +2 \text{ kN}$$

$$\sigma_{max} = \frac{F_{max}}{A}$$

* Nó A:



$$\Sigma F_x = 0 \rightarrow F_{AB} \cdot \sin 71 + 70 - F_{AD} \cdot \sin 71 = 0$$

* Em continuidade as reações pelas outras nós

* Nó D:

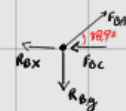


$$\Sigma F_y = 0 \rightarrow F_{DA} \cdot \sin 18,9 = 24$$

$$F_{DA} = \frac{24}{\sin 18,9} = 72,85 \text{ kN}$$

$$\Sigma F_x = 0 \rightarrow F_{DC} = +0 \text{ kN}$$

* Nó B:



$$\Sigma F_y = 0 \rightarrow F_{BA} \cdot \sin 71 = 48 \text{ kN} \rightarrow F_{BA} = 78,78 \text{ kN}$$

$$\Sigma F_x = 0 \rightarrow F_{BA} \cdot \cos 71 + F_{BC} - R_{Bx} = 0 \rightarrow F_{BC} = 70$$

* Nó C:



$$\Sigma F_x = 0 \rightarrow F_{CB} = F_{CD}$$

$$F_{CB} = 70 \text{ kN}$$

$$\Sigma F_y = 0 \rightarrow R_{Cy} - F_{AC} = 0$$

$$F_{AC} = 2 \text{ kN}$$

$$\sigma = \frac{78,78 \cdot 10^3}{903,902} = 87,186666,67 \text{ Pa}$$